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Sliding component

Abstract

There is provided a sliding component that allows sliding surfaces to smoothly slide against each other from a low-speed state to a high-speed state in both forward and reverse relative rotation directions. A deep groove is provided that partitions a fluid-side region where a fluid-side groove and a fluid-side reverse groove are provided and a leakage-side region where leakage-side grooves and leakage-side reverse grooves are provided off from each other.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a sliding component, for example, a sliding component used for a shaft seal or a bearing.

BACKGROUND ART

(2) As a sliding component that prevents a leakage of a sealed fluid around a rotating shaft of a rotating machine, for example, there is known a mechanical seal including a pair of sliding rings having an annular shape which rotate relative to each other and of which sliding surfaces slide against each other. In such a mechanical seal, in recent years, a reduction in energy loss caused by sliding has been desired for environmental measures and the like, and dynamic pressure generation grooves may be provided on a sliding surface of a sliding ring.

(3) For example, in a mechanical seal disclosed in Patent Citation 1, a plurality of spiral grooves on a radially inner side and a plurality of spiral grooves on a radially outer side are provided on a sliding surface of one sliding ring. The spiral grooves on the radially inner side communicate with a space on a leakage side that is the radially inner side of the sliding surface, and extend in a radially outward direction while inclining in one circumferential direction. In addition, the spiral grooves on the radially outer side communicate with a space on a sealed fluid side that is the radially outer side of the sliding surface, and extend in a radially inward direction while inclining in the other circumferential direction. Since the spiral grooves extend in a radial direction while inclining in the circumferential directions in such a manner, a higher dynamic pressure generation capability can be obtained compared to recessed portions or grooves such as Rayleigh steps extending along the radial direction or dimples.

(4) When the sliding rings rotate relative to each other at low speed, a sealed fluid flows into a gap between the sliding surfaces from the spiral grooves on the radially outer side, and a liquid film is formed. In addition, when the sliding rings rotate relative to each other at high speed, a gas on the leakage side is suctioned from starting end portions on the radially inner side of the spiral grooves on the radially inner side, and a dynamic pressure is generated at terminating end portions, so that the sliding surfaces are slightly separated from each other, and the lubricity between the sliding surfaces is improved. In addition, when the sliding rings rotate relative to each other at high speed, the spiral grooves on the radially outer side suction the sealed fluid between the sliding surfaces, and discharge the sealed fluid to the radially outer side, so that the leakage of the sealed fluid into the space on the leakage side is suppressed.

CITATION LIST

Patent Literature

(5) Patent Citation 1: WO 2018/051867 A (Page 10, FIG. 2)

SUMMARY OF INVENTION

Technical Problem

(6) Depending on the type of the rotating machine, the rotation direction may be switched according to the situation, and a mechanical seal capable of coping with both the relative rotation directions of the sliding rings has been desired. However, the mechanical seal as disclosed in Patent Citation 1 can be used when the sliding rings rotate relative to each other at low speed in a reverse direction, but cannot actually be used during high-speed rotation, and cannot cope with an operation from a low-speed state to a high-speed state in both the relative rotation directions of the

sliding rings.
(7) The present invention has been made in view of such problems, and an object of the present invention is to provide a sliding component that allows sliding surfaces to smoothly slide against each other from a low-speed state to a high-speed state in both forward and reverse relative rotation directions.
Solution to Problem
(8) In order to solve the foregoing problems, according to the present invention, there is provided a sliding component including a pair of sliding surfaces disposed in a rotating machine to face each other at a location where the pair of sliding surfaces rotate relative to each other when the rotating machine is driven, and partitioning a fluid-side space and a leakage-side space off from each other, wherein one of the sliding surfaces is provided with: a fluid-side groove communicating with the fluid-side space and extending in a forward direction of relative rotation of the one of the sliding surfaces; a fluid-side reverse groove communicating with the fluid-side space and extending in a reverse direction of the relative rotation; a leakage-side groove of which at least one end portion is disposed closer to a side of the leakage-side space than the fluid-side groove and the fluid-side reverse groove, and which extends from the end portion toward the fluid-side space in the forward direction of the relative rotation; a leakage-side reverse groove of which at least one end portion is disposed closer to the side of the leakage-side space than the fluid-side groove and the fluid-side reverse groove, and which extends from the end portion toward the fluid-side space in the reverse direction of the relative rotation; and a deep groove that partitions a fluid-side region where the fluid-side groove and the fluid-side reverse groove are provided and a leakage-side region where the leakage-side groove and the leakage-side reverse groove are provided off from each other. According to the aforesaid feature of the present invention, when the relative rotation of the sliding component is low-speed forward rotation, the lubricity between the sliding surfaces is improved mainly by a sealed fluid supplied from a closed end portion of the fluid-side groove into a gap between the sliding surfaces. During high-speed forward rotation, since the sealed fluid supplied from the fluid-side groove into the gap between the sliding surfaces is collected in the deep groove, a positive pressure in the fluid-side groove is suppressed. In addition, when the relative rotation is low-speed reverse rotation, the lubricity between the sliding surfaces is improved mainly by the sealed fluid supplied from a closed end portion of the fluid-side reverse groove into the gap between the sliding surfaces. During high-speed reverse rotation, since the sealed fluid supplied from the fluid-side reverse groove into the gap between the sliding surfaces is collected in the deep groove, a positive pressure in the fluid-side reverse groove is suppressed. In addition, since the fluid that has flowed into the gap between the sliding surfaces from the closed end portion of one fluid-side groove or the fluid-side reverse groove is collected by the other fluid-side groove or the fluid-side reverse groove and the deep groove, the leakage of the sealed fluid into the leakage-side space due to the suction function of the leakage-side groove or the leakage-side reverse groove is suppressed.
(9) It may be preferable that the deep groove communicates with the fluid-side space. According to this preferable configuration, since the fluid from the fluid-side space flows in and out from the deep groove, the amount of the sealed fluid in the deep groove is stabilized.
(10) It may be preferable that each of leakage-side end portions of the leakage-side groove and the leakage-side reverse groove communicates with the leakage-side space. According to this preferable configuration, during relative rotation in the forward direction, a leakage-side fluid can be efficiently introduced from the leakage-side space into the leakage-side groove, and during relative rotation in the reverse direction, the leakage-side fluid can be efficiently introduced from the leakage-side space into the leakage-side reverse groove.
(11) It may be preferable that the one of the sliding surfaces is provided with a communication groove communicating with each of leakage-side end portions of the leakage-side groove and the leakage-side reverse groove. According to this preferable configuration, during relative rotation in

the forward direction, the fluid collected in the leakage-side reverse groove can be guided to the leakage-side groove through the communication groove, and during relative rotation in the reverse direction, the fluid collected in the leakage-side groove can be guided to the leakage-side reverse groove through the communication groove.
(12) It may be preferable that the communication groove has an annular shape. According to this preferable configuration, the fluid suctioned into the leakage-side groove or the leakage-side reverse groove can be collected in the communication groove having an annular shape, and the fluid in the communication groove having an annular shape can be guided to the leakage-side groove or the leakage-side reverse groove. Further, since a land is formed on a leakage side of the annular deep groove, the fluid collected in the communication groove is less likely to leak into the leakage-side space.
(13) It may be preferable that each of terminating end portions of the leakage-side groove and the leakage-side reverse groove is disposed closer to a fluid-side space side than each of terminating end portions of the fluid-side groove and the fluid-side reverse groove. According to this preferable configuration, a sufficient length can be ensured for each of the grooves.
(14) It may be preferable that the fluid-side groove, the leakage-side reverse groove, the fluid-side reverse groove, and the leakage-side groove are disposed in an inclined manner, and the deep groove includes a first inclined portion extending along the fluid-side groove and the leakage-side reverse groove, and a second inclined portion extending along the fluid-side reverse groove and the leakage-side groove. According to this preferable configuration, since the first inclined portion can be disposed between the fluid-side groove and the leakage-side reverse groove, and the second inclined portion can be disposed between the fluid-side reverse groove and the leakage-side groove, a plurality of the grooves can be efficiently disposed in a circumferential direction.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a longitudinal sectional view illustrating one example of a mechanical seal as a sliding component according to a first embodiment of the present invention.
(2) FIG. **2** is a view of a sliding surface of a stationary seal ring in the first embodiment as viewed in an axial direction.
(3) FIG. **3** is an enlarged view of the sliding surface of the stationary seal ring when a rotating seal ring rotates forward at low speed in the first embodiment as viewed in the axial direction.
(4) FIG. **4** is an enlarged view of the sliding surface of the stationary seal ring when the rotating seal ring rotates forward at high speed in the first embodiment as viewed in the axial direction.
(5) FIG. **5** is a view of a sliding surface of a stationary seal ring included in a sliding component according to a second embodiment of the present invention as viewed in the axial direction.
(6) FIG. **6** is a view of a sliding surface of a stationary seal ring included in a sliding component according to a third embodiment of the present invention as viewed in the axial direction.
(7) FIG. **7** is a view of a sliding surface of a stationary seal ring included in a sliding component according to a fourth embodiment of the present invention as viewed in the axial direction.
(8) FIG. **8** is a view of a sliding surface of a stationary seal ring included in a sliding component according to a fifth embodiment of the present invention as viewed in the axial direction.
(9) FIG. **9** is a schematic view illustrating a modification example of fluid-side grooves and fluid-side reverse grooves in the first to fifth embodiments.
DESCRIPTION OF EMBODIMENTS
(10) Modes for implementing a sliding component according to the present invention will be described below based on embodiments.
First Embodiment

(11) A sliding component according to a first embodiment of the present invention will be described with reference to FIGS. **1** to **4**. Incidentally, in the present embodiment, a mechanical seal will be described as an example of the sliding component. The mechanical seal of the present embodiment will be described as being configured such that atmosphere A exists in an inner space S**1**, a sealed fluid F exists in an outer space S**2**, a radially inner side of sliding rings constituting the mechanical seal is a leakage side (low-pressure side), and a radially outer side is a sealed fluid side (high-pressure side). In addition, for convenience of description, in the drawings, dots may be added to grooves and the like formed on a sliding surface.
(12) The mechanical seal illustrated in FIG. **1** is an inside mechanical seal that seals the sealed fluid F in the outer space S**2** as a fluid-side space, which tends to leak from the radially outer side toward the radially inner side of sliding surfaces, and that allows the inner space S**1** as a leakage-side space to communicate with the atmosphere A. Incidentally, in the present embodiment, a mode in which the sealed fluid F is a high-pressure liquid and the atmosphere A is a gas having a lower pressure than the sealed fluid F will be provided as an example.
(13) The mechanical seal mainly includes a stationary seal ring **10** and a rotating seal ring **20**. The stationary seal ring **10** has an annular shape, and is provided on a seal cover **5** fixed to a housing **4** of an attached device, so as to be non-rotatable and movable in an axial direction. The rotating seal ring **20** has an annular shape, and is provided on a rotating shaft **1** via a sleeve **2** so as to be rotatable together with the rotating shaft **1**. The stationary seal ring **10** is biased in the axial direction by an elastic member **7**, so that a sliding surface **11** of the stationary seal ring **10** and a sliding surface **21** of the rotating seal ring **20** slide in close contact with each other. Incidentally, the sliding surface **21** of the rotating seal ring **20** is a flat surface, and a recessed portion such as a groove is not provided on the flat surface.
(14) The stationary seal ring **10** and the rotating seal ring **20** are typically made of SiC (as an example of hard material) or a combination of SiC and carbon (as an example of soft material); however, the present invention is not limited to these materials, and any sliding material that is used as a sliding material for mechanical seals can be applied. Incidentally, examples of SiC include sintered bodies using boron, aluminum, carbon, or the like as a sintering aid, and materials consisting of two or more phases with different components and compositions, such as SiC in which graphite particles are dispersed, reaction-sintered SiC consisting of SiC and Si, SiC—TiC, and SiC—TiN. As carbon, mixed carbon of a carbonaceous substance and a graphitic substance, resin-molded carbon, sintered carbon, and the like can be used. In addition, in addition to the above-described sliding materials, metal materials, resin materials, surface modification materials (e.g., coating materials), composite materials, and the like can also be applied.
(15) As illustrated in FIG. **2**, the rotating seal ring **20** that is a mating seal ring slides relative to the stationary seal ring **10** clockwise as indicated by a solid arrow or counterclockwise as indicated by a dashed arrow. Hereinafter, the direction of the solid arrow will be described as a forward rotation direction of the rotating seal ring **20**, and the direction of the dashed arrow will be described as a reverse rotation direction of the rotating seal ring **20**.
(16) The sliding surface **11** of the stationary seal ring **10** is provided with a plurality of fluid-side spiral grooves **13** as fluid-side grooves; a plurality of fluid-side reverse spiral grooves **14** as fluid-side reverse grooves; a plurality of leakage-side spiral grooves **15***a* to **15***c* as leakage-side grooves; a plurality of leakage-side reverse spiral grooves **16***a* to **16***c* as leakage-side reverse grooves; and a plurality of deep grooves **17**. Here, in this specification, the spiral grooves are configured such that each extension direction of the grooves has both a radial component and a circumferential component.
(17) The plurality (three in the present embodiment) of fluid-side spiral grooves **13** are arranged in a circumferential direction on the radially outer side of the sliding surface **11**. The fluid-side spiral grooves **13** communicate with the outer space S**2** at end portions **13**A on the radially outer side of the fluid-side spiral grooves **13**, and extend in the forward rotation direction of the rotating seal

ring **20**, namely, in one circumferential direction with reference to communication locations thereof.

(18) In detail, the fluid-side spiral grooves **13** are spiral grooves extending in an arcuate shape while inclining with a clockwise component from the radially outer side toward the radially inner side. In addition, end portions **13**B on the radially inner side of the fluid-side spiral grooves **13** have a closed shape, namely, are closed end portions. The fluid-side spiral grooves **13** have a constant depth in the extension direction.

(19) The plurality (three in the present embodiment) of fluid-side reverse spiral grooves **14** are arranged in the circumferential direction on the radially outer side of the sliding surface **11**. The fluid-side reverse spiral grooves **14** communicate with the outer space S**2** at end portions **14**A on the radially outer side of the fluid-side reverse spiral grooves **14**, and extend in the reverse rotation direction of the rotating seal ring **20**, namely, in the other circumferential direction with reference to communication locations thereof.

(20) In detail, the fluid-side reverse spiral grooves **14** are spiral grooves extending in an arcuate shape while inclining with a counterclockwise component from the radially outer side toward the radially inner side. In addition, end portions **14**B on the radially inner side of the fluid-side reverse spiral grooves **14** have a closed shape, namely, are closed end portions. The fluid-side reverse spiral grooves **14** have a constant depth in the extension direction. Incidentally, in the present embodiment, the fluid-side spiral grooves **13** and the fluid-side reverse spiral grooves **14** have the same depth. In other words, the fluid-side reverse spiral grooves **14** have a symmetrical shape with respect to the fluid-side spiral grooves **13** in the circumferential direction.

(21) The plurality of each (three each in the present embodiment) of the leakage-side spiral grooves **15***a* to **15***c* are arranged in the circumferential direction on the radially inner side of the sliding surface **11**. The leakage-side spiral grooves **15***a* to **15***c* communicate with the inner space S**1** at leakage-side end portions of the leakage-side spiral grooves **15***a* to **15***c*, namely, end portions **15**A on the radially inner side, and extend in the forward rotation direction of the rotating seal ring **20**, namely, in the one circumferential direction with reference to communication locations thereof. The leakage-side spiral grooves **15***a* to **15***c* extend substantially parallel to the fluid-side spiral grooves **13**.

(22) In detail, the leakage-side spiral grooves **15***a* to **15***c* are spiral grooves extending in an arcuate shape while inclining with a clockwise component from the radially inner side toward the radially outer side. In addition, end portions **15**B on the radially outer side of the leakage-side spiral grooves **15***a* to **15***c* have a closed shape, namely, are closed end portions. The leakage-side spiral grooves **15***a* to **15***c* have a constant depth in the extension direction.

(23) In addition, the three leakage-side spiral grooves **15***a* to **15***c* have different lengths. The leakage-side spiral groove **15***a* is longer than the leakage-side spiral groove **15***b*, and the leakage-side spiral groove **15***b* is longer than the leakage-side spiral groove **15***c*. The end portions **15**B of the leakage-side spiral grooves **15***a* to **15***c* are disposed side by side in the radial direction.

(24) In addition, the end portions **15**B of the leakage-side spiral grooves **15***a* to **15***c* are disposed closer to the radially outer side than the end portions **13**B of the fluid-side spiral grooves **13** and the end portions **14**B of the fluid-side reverse spiral grooves **14**.

(25) The plurality of each (three each in the present embodiment) of the leakage-side reverse spiral grooves **16***a* to **16***c* are arranged in the circumferential direction on the radially inner side of the sliding surface **11**. The leakage-side reverse spiral grooves **16***a* to **16***c* communicate with the inner space S**1** at leakage-side end portions of the leakage-side reverse spiral grooves **16***a* to **16***c*, namely, end portions **16**A on the radially inner side, and extend in the reverse rotation direction of the rotating seal ring **20**, namely, in the other circumferential direction with reference to communication locations thereof. The leakage-side reverse spiral grooves **16***a* to **16***c* extend substantially parallel to the fluid-side reverse spiral grooves **14**.

(26) In detail, the leakage-side reverse spiral grooves **16***a* to **16***c* are spiral grooves extending in an

arcuate shape while inclining with a counterclockwise component from the radially inner side toward the radially outer side. In addition, end portions **16**B on the radially outer side of the leakage-side reverse spiral grooves **16***a* to **16***c* have a closed shape, namely, are closed end portions. In other words, the leakage-side reverse spiral grooves **16***a* to **16***c* have a symmetrical shape with respect to the leakage-side spiral grooves **15***a* to **15***c* in the circumferential direction.
(27) The leakage-side reverse spiral grooves **16***a* to **16***c* have a constant depth in the extension direction. Incidentally, in the present embodiment, the leakage-side spiral grooves **15***a* to **15***c* and the leakage-side reverse spiral grooves **16***a* to **16***c* have the same depth, and the leakage-side spiral grooves **15***a* to **15***c* and the leakage-side reverse spiral grooves **16***a* to **16***c* are formed deeper than the fluid-side spiral grooves **13** and the fluid-side reverse spiral grooves **14**. Further, the capacities of the leakage-side spiral grooves **15***a* to **15***c* and the leakage-side reverse spiral grooves **16***a* to **16***c* are larger than the capacities of the fluid-side spiral grooves **13** and the fluid-side reverse spiral grooves **14**.
(28) The leakage-side reverse spiral grooves **16***a* to **16***c* have different lengths. The leakage-side reverse spiral groove **16***a* is longer than the leakage-side reverse spiral groove **16***b*, and the leakage-side reverse spiral groove **16***b* is longer than the leakage-side reverse spiral groove **16***c*. The end portions **16**B of the leakage-side reverse spiral grooves **16***a* to **16***c* are disposed side by side in the radial direction.
(29) In addition, the end portions **16**B of the leakage-side reverse spiral grooves **16***a* to **16***c* are disposed closer to the radially outer side than the end portions **13**B of the fluid-side spiral grooves **13** and the end portions **14**B of the fluid-side reverse spiral grooves **14**.
(30) Incidentally, hereinafter, one set of the fluid-side spiral groove **13** and the fluid-side reverse spiral groove **14** facing each other such that the end portions **13**B and **14**B approach each other in the circumferential direction will be described, and one set of the leakage-side spiral grooves **15***a* to **15***c* and the leakage-side reverse spiral grooves **16***a* to **16***c* facing each other such that the end portions **15**B and **16**B approach each other in the circumferential direction will be described.
(31) One set of the leakage-side spiral grooves **15***a* to **15***c* and the leakage-side reverse spiral grooves **16***a* to **16***c* are arranged between the fluid-side spiral groove **13** of one set and the fluid-side reverse spiral groove **14** of the other adjacent set in the circumferential direction.
(32) The plurality (three in the present embodiment) of deep grooves **17** are arranged in the circumferential direction on the sliding surface **11**. The deep grooves **17** have a constant depth in the extension direction. The deep grooves **17** are formed deeper than the leakage-side spiral grooves **15***a* to **15***c* and the leakage-side reverse spiral grooves **16***a* to **16***c*. Incidentally, the depth of the deep grooves **17** is set to such a depth that almost no dynamic pressure is generated by relative rotation between the stationary seal ring **10** and the rotating seal ring **20**.
(33) The deep groove **17** communicates with the outer space S**2** at both end portions **17**A and **17**B in the circumferential direction, and extends to surround one set of the fluid-side spiral grooves **13** and the fluid-side reverse spiral grooves **14**. In detail, the deep groove **17** includes a first portion **171** as a first inclined portion; a second portion **172** as a second inclined portion; and a third portion **173**. The first portion **171** is a portion extending substantially parallel to the fluid-side spiral groove **13** and the leakage-side reverse spiral grooves **16***a* to **16***c* therebetween. The second portion **172** is a portion extending substantially parallel to the fluid-side reverse spiral groove **14** and the leakage-side spiral grooves **15***a* to **15***c* therebetween. The third portion **173** is a portion extending concentrically with the stationary seal ring **10**, and connecting radially inner ends of the first portion **171** and the second portion **172**.
(34) In addition, portions of the sliding surface **11** other than the fluid-side spiral groove **13**, the fluid-side reverse spiral groove **14**, the leakage-side spiral grooves **15***a* to **15***c*, the leakage-side reverse spiral grooves **16***a* to **16***c*, and the deep groove **17** are lands **12** having flat surfaces disposed on the same plane. The flat surfaces of the lands **12** function as sliding surfaces that substantially slide against the sliding surface **21** of the rotating seal ring **20**.

(35) The sliding surface **11** is partitioned into fluid-side regions A**1** and a leakage-side region A**2** by the deep grooves **17**. In each of the fluid-side regions A**1**, one set of the fluid-side spiral groove **13** and the fluid-side reverse spiral groove **14** and the land **12** that partitions the fluid-side spiral groove **13** and the fluid-side reverse spiral groove **14** off from each other are provided. In the leakage-side region A**2**, all the leakage-side spiral grooves **15***a* to **15***c* and the leakage-side reverse spiral grooves **16***a* to **16***c*, and the land **12** that partitions the leakage-side spiral grooves **15***a* to **15***c* and the leakage-side reverse spiral grooves **16***a* to **16***c* off from each other are provided. In other words, the leakage-side region A**2** is a region of the sliding surface **11** other than the fluid-side regions A**1**.
(36) Next, the flow of the sealed fluid F and the atmosphere A during relative rotation between the stationary seal ring **10** and the rotating seal ring **20** will be schematically described with reference to FIG. **2**. Incidentally, here, a description will be provided without specifying a relative rotation speed between the stationary seal ring **10** and the rotating seal ring **20**.
(37) When the rotating seal ring **20** rotates relative to the stationary seal ring **10** in the forward direction, in the fluid-side spiral groove **13**, the sealed fluid F moves to the end portion **13**B, and a positive pressure is generated at the end portion **13**B and in the vicinity of the end portion **13**B. In addition, in the fluid-side reverse spiral groove **14**, the sealed fluid F moves to the end portion **14**A, and a relative negative pressure is generated at the end portion **14**B and in the vicinity of the end portion **14**B. In addition, in the leakage-side spiral grooves **15***a* to **15***c*, the atmosphere A moves to each of the end portions **15**B, and a positive pressure is generated at each of the end portions **15**B and in the vicinity of each of the end portions **15**B. In addition, in the leakage-side reverse spiral grooves **16***a* to **16***c*, the atmosphere A moves to each of the end portions **16**A, and a relative negative pressure is generated at each of the end portions **16**B and in the vicinity of each of the end portions **16**B. Incidentally, the relative negative pressure referred to here represents a state where the pressure is lower than the surrounding pressure, rather than a vacuum state.
(38) On the other hand, when the rotating seal ring **20** rotates relative to the stationary seal ring **10** in the reverse direction, in the fluid-side spiral groove **13**, the sealed fluid F moves to the end portion **13**A, and a relative negative pressure is generated at the end portion **13**B and in the vicinity of the end portion **13**B. In addition, in the fluid-side reverse spiral groove **14**, the sealed fluid F moves to the end portion **14**B, and a positive pressure is generated at the end portion **14**B and in the vicinity of the end portion **14**B. In addition, in the leakage-side spiral grooves **15***a* to **15***c*, the atmosphere A moves to each of the end portions **15**A, and a relative negative pressure is generated at each of the end portions **15**B and in the vicinity of each of the end portions **15**B. In addition, in the leakage-side reverse spiral grooves **16***a* to **16***c*, the atmosphere A moves to each of the end portions **16**B, and a positive pressure is generated at each of the end portions **16**B and in the vicinity of each of the end portions **16**B.
(39) Next, a change in the force that separates the sliding surfaces **11** and **21** will be described with reference to FIGS. **3** and **4**. Incidentally, here, a case where the rotating seal ring **20** rotates in the forward rotation direction will be described as an example, and the description of a case where the rotating seal ring **20** in the reverse rotation direction will be omitted. In addition, in FIGS. **3** and **4**, for convenience of description, the range of a positive pressure generated in each of the spiral grooves is illustrated as being surrounded by an alternating long and short dashed line.
(40) First, when the rotating seal ring **20** is not in rotation and is stopped, the sealed fluid F flows into the fluid-side spiral groove **13** and the fluid-side reverse spiral groove **14** from openings of the end portions **13**A and **14**A. In addition, the sealed fluid F flows into the deep groove **17** from openings of the end portions **17**A and **17**B. In addition, the atmosphere A flows into the leakage-side spiral grooves **15***a* to **15***c* and the leakage-side reverse spiral grooves **16***a* to **16***c* from openings of the end portions **15**A and **16**A. Incidentally, since the stationary seal ring **10** is biased to a rotating seal ring **20** side by the elastic member **7**, the sliding surfaces **11** and **21** are in contact with each other, and there is almost no leakage of the sealed fluid F between the sliding surfaces **11**

and **21** into the inner space S**1**.

(41) At low speed immediately after the rotating seal ring **20** starts rotating relative to the stationary seal ring **10**, as illustrated in FIG. **3**, a positive pressure is generated at the end portion **13**B of the fluid-side spiral groove **13** and each of the end portions **15**B of the leakage-side spiral grooves **15***a* to **15***c*. The sliding surfaces **11** and **21** are slightly separated from each other by the positive pressure.

(42) In detail, since the sealed fluid F having a higher pressure than the atmosphere A flowing into the leakage-side spiral grooves **15***a* to **15***c* flows into the fluid-side spiral groove **13**, and the capacity of the fluid-side spiral groove **13** is smaller than that of the leakage-side spiral grooves **15***a* to **15***c*, a first force caused by the positive pressure generated at the end portion **13**B of the fluid-side spiral groove **13** is larger than a second force caused by the positive pressure generated at each of the end portions **15**B of the leakage-side spiral grooves **15***a* to **15***c*. For this reason, during low-speed rotation of the rotating seal ring **20**, the first force mainly acts to separate the sliding surfaces **11** and **21** from each other.

(43) The sealed fluid F is supplied from the fluid-side spiral groove **13** into a gap between the sliding surfaces **11** and **21** in such a manner, and the sliding surfaces **11** and **21** are slightly separated from each other, so that even during low-speed rotation, lubricity can be improved and wear between the sliding surfaces **11** and **21** can be suppressed.

(44) In addition, the sealed fluid F supplied to the fluid-side region A**1** on the sliding surface **11** is mainly suctioned into the fluid-side reverse spiral groove **14** as indicated by arrow H**1**, and is partially collected in the deep groove **17** as indicated by arrow H**2**. In addition, since the floating distance between the sliding surfaces **11** and **21** is slight, almost no sealed fluid F flows into the leakage-side region A**2** on the sliding surface **11**. For this reason, the leakage of the sealed fluid F into the inner space S**1** is suppressed. Incidentally, when a larger amount of the sealed fluid F than the amount of the sealed fluid F that can be stored in the deep groove **17** is collected, the excess of the sealed fluid F is returned to the outer space S**2**.

(45) When the relative rotation speed of the rotating seal ring **20** increases, as illustrated in FIG. **4**, the positive pressure generated at the end portion **13**B of the fluid-side spiral groove **13** and each of the end portions **15**B of the leakage-side spiral grooves **15***a* to **15***c* gradually increases. Accordingly, the first force and the second force are increased, so that the sliding surfaces **11** and **21** are further separated from each other compared to the state of FIG. **3**.

(46) Incidentally, when the relative rotation speed of the rotating seal ring **20** reaches a certain level or higher, since some of the sealed fluid F supplied from the end portion **13**B of the fluid-side spiral groove **13** into the gap between the sliding surfaces **11** and **21** is collected in the deep groove **17** as indicated by arrow H**3**, the first force caused by the positive pressure generated at the end portion **13**B of the fluid-side spiral groove **13** is prevented from increasing any further.

(47) When the relative rotation speed of the rotating seal ring **20** further increases and reaches high-speed rotation, namely, a steady-state operating state, the inflow amount of the atmosphere A suctioned into the leakage-side spiral grooves **15***a* to **15***c* further increases so that a high positive pressure is generated to increase the second force. Therefore, the sliding surfaces **11** and **21** are more significantly separated from each other.

(48) In the present embodiment, when the floating distance increases due to high-speed rotation of the rotating seal ring **20**, the positive pressure generated in the fluid-side spiral groove **13** becomes negligibly small. Therefore, during high-speed rotation of the rotating seal ring **20**, the second force mainly acts to separate the sliding surfaces **11** and **21** from each other.

(49) In addition, in a steady-state operating state of the mechanical seal, the negative pressure generated at each of the end portions **16**B of the leakage-side reverse spiral grooves **16***a* to **16***c* and in the vicinity of each of the end portions **16**B is relatively small, and the positive pressure generated at each of the end portions **15**B of the facing leakage-side spiral grooves **15***a* to **15***c* acts to push most of the sealed fluid F flowing in from the radially outer side of the leakage-side region

A**2**, out into the outer space S**2**.

(50) In such a manner, when the relative rotation of the mechanical seal is low-speed forward rotation, the lubricity between the sliding surfaces **11** and **21** is improved mainly by the sealed fluid F supplied from the end portion **13**B of the fluid-side spiral groove **13** into the gap between the sliding surfaces **11** and **21**. In addition, when the relative rotation of the mechanical seal is high-speed forward rotation, since the sealed fluid F supplied from the fluid-side spiral groove **13** into the gap between the sliding surfaces **11** and **21** is collected in the deep groove **17**, the positive pressure in the fluid-side spiral groove **13** is suppressed, and the sliding surfaces **11** and **21** are separated from each other mainly by the positive pressure generated at each of the end portions **15**B of the leakage-side spiral grooves **15***a* to **15***c*, so that the lubricity is improved.

(51) In addition, since the sealed fluid F that has flowed into the gap between the sliding surfaces **11** and **21** from the end portion **13**B of the fluid-side spiral groove **13** is collected by the fluid-side reverse spiral groove **14** and the deep groove **17**, the leakage of the sealed fluid F into the inner space S**1** due to the suction function of the leakage-side reverse spiral grooves **16***a* to **16***c* is suppressed.

(52) In addition, since the sliding surface **11** is provided with the fluid-side reverse spiral groove **14** having a symmetrical shape with respect to the fluid-side spiral groove **13** in the circumferential direction, and with the leakage-side reverse spiral grooves **16***a* to **16***c* having a symmetrical shape with respect to the leakage-side spiral grooves **15***a* to **15***c* in the circumferential direction, when the relative rotation of the mechanical seal is reverse rotation, the same function as described above is exhibited.

(53) In addition, since the deep groove **17** communicates with the outer space S**2**, the inflow and outflow of the sealed fluid F can be performed between the deep groove **17** and the outer space S**2**, and the amount of the sealed fluid F in the deep groove **17** is stabilized. Namely, the sealed fluid F supplied from the fluid-side spiral groove **13** or the fluid-side reverse spiral groove **14** into the gap between the sliding surfaces **11** and **21** can be reliably collected in the deep groove **17**.

(54) In addition, each of the end portions **15**A and **16**A of the leakage-side spiral grooves **15***a* to **15***c* and the leakage-side reverse spiral grooves **16***a* to **16***c* communicates with the inner space S**1**. According to this configuration, during relative rotation of the mechanical seal in the forward direction, the atmosphere A can be introduced from the inner space S**1** into the leakage-side spiral grooves **15***a* to **15***c*, and during relative rotation in the reverse direction, the atmosphere A can be introduced from the inner space S**1** into the leakage-side reverse spiral grooves **16***a* to **16***c*, so that a positive pressure can be stably generated in the leakage-side spiral grooves **15***a* to **15***c* and the leakage-side reverse spiral grooves **16***a* to **16***c*.

(55) In addition, each of the end portions **15**B of the leakage-side spiral grooves **15***a* to **15***c* and each of the end portions **16**B of the leakage-side reverse spiral grooves **16***a* to **16***c* are disposed closer to the radially outer side, namely, an outer space S**2** side than the end portion **13**B of the fluid-side spiral groove **13** and the end portion **14**B of the fluid-side reverse spiral groove **14**. According to this configuration, a sufficient length can be ensured for each of the spiral grooves.

(56) In addition, since the deep groove **17** includes the first portion **171** extending along the fluid-side spiral groove **13** and the leakage-side reverse spiral grooves **16***a* to **16***c* and the second portion **172** extending along the fluid-side reverse spiral groove **14** and the leakage-side spiral grooves **15***a* to **15***c*, the plurality of spiral grooves can be efficiently disposed in the circumferential direction.

Second Embodiment

(57) Next, a sliding component according to a second embodiment of the present invention will be described with reference to FIG. **5**. Incidentally, the description of configurations that are the same as and overlap with the configurations of the first embodiment will be omitted.

(58) An annular land **18** is provided at an edge portion on the radially inner side of a sliding surface **111** of a stationary seal ring **100** of the second embodiment. An annular deep groove **117** as a communication groove is formed concentrically with the stationary seal ring **100** on the radially

outer side of the annular land **18**. The annular deep groove **117** and the third portion **173** of the deep groove **17** are partitioned off from each other by a land **112** forming a leakage-side region A**20**. Incidentally, the annular deep groove **117** is formed to such a depth that almost no dynamic pressure is generated by relative rotation of the mechanical seal.
(59) Leakage-side spiral grooves **151***a* to **151***c* and leakage-side reverse spiral grooves **161***a* to **161***c* are provided on the radially outer side of the annular deep groove **117**. Each of end portions **151**A of the leakage-side spiral grooves **151***a* to **151***c* and each of end portions **161**A of the leakage-side reverse spiral grooves **161***a* to **161***c* communicate with the annular deep groove **117**. Incidentally, configurations other than described above are the same as the configurations of the first embodiment.
(60) According to this configuration, during relative rotation of the mechanical seal in the forward direction, the sealed fluid F collected in the leakage-side reverse spiral grooves **161***a* to **161***c* can be guided to the leakage-side spiral grooves **151***a* to **151***c* through the annular deep groove **117**. In addition, during relative rotation of the mechanical seal in the reverse direction, the sealed fluid F collected in the leakage-side spiral grooves **151***a* to **151***c* can be guided to the leakage-side reverse spiral grooves **161***a* to **161***c* through the annular deep groove **117**. For this reason, the sealed fluid F collected in the leakage-side spiral grooves **151***a* to **151***c* and the leakage-side reverse spiral grooves **161***a* to **161***c* is less likely to leak into the inner space S**1**.
(61) In addition, the annular deep groove **117** can collect the sealed fluid F from a portion other than the leakage-side spiral grooves **151***a* to **151***c* and the leakage-side reverse spiral grooves **161***a* to **161***c*, namely, from the land **112** forming the leakage-side region A**20**.
(62) Further, the annular land **18** is formed on the radially inner side of the annular deep groove **117**. In other words, since the annular deep groove **117** and the inner space S**1** are divided by the annular land **18**, the sealed fluid F collected in the annular deep groove **117** is less likely to leak into the inner space S**1**.
(63) Incidentally, in the present embodiment, the mode in which the deep groove and the communication groove having an annular shape are partitioned off from each other by the land has been provided as an example; however, the deep groove and the communication groove having an annular shape may communicate with each other.
Third Embodiment
(64) Next, a sliding component according to a third embodiment of the present invention will be described with reference to FIG. **6**. Incidentally, the description of configurations that are the same as and overlap with the configurations of the second embodiment will be omitted.
(65) Two fluid-side spiral grooves **213***a* and **213***b* and two fluid-side reverse spiral grooves **214***a* and **214***b* are provided in a fluid-side region A**10** of a sliding surface **211** of a stationary seal ring **200** of the third embodiment.
(66) The fluid-side spiral groove **213***a* is longer than the fluid-side spiral groove **213***b*. In addition, the fluid-side spiral groove **213***a* is disposed side by side with the fluid-side spiral groove **213***b* in the reverse rotation direction of the rotating seal ring **20**.
(67) The fluid-side reverse spiral groove **214***a* is longer than the fluid-side reverse spiral groove **214***b*, and has the same length as the fluid-side spiral groove **213***a*. In addition, the fluid-side reverse spiral groove **214***a* is disposed side by side with the fluid-side reverse spiral groove **214***b* in the forward rotation direction of the rotating seal ring **20**.
(68) In addition, the fluid-side spiral groove **213***b* and the fluid-side reverse spiral groove **214***b* have the same length. Incidentally, configurations other than described above are the same as the configurations of the second embodiment.
Fourth Embodiment
(69) Next, a sliding component according to a fourth embodiment of the present invention will be described with reference to FIG. **7**. Incidentally, the description of configurations that are the same as and overlap with the configurations of the second embodiment will be omitted.

(70) A sliding surface **311** of a stationary seal ring **300** of the fourth embodiment is provided with a deep groove **317** that is continuous in the circumferential direction. The deep groove **317** includes a first portion **317***a*, a second portion **317***b*, a third portion **317***c*, and a fourth portion **317***d*.
(71) The first portion **317***a* is a portion extending substantially parallel to the fluid-side spiral groove **13** and the leakage-side reverse spiral grooves **161***a* to **161***c* therebetween. A radially outer end of the first portion **317***a* does not communicate with the outer space S**2**. The second portion **317***b* is a portion extending substantially parallel to the fluid-side reverse spiral groove **14** and the leakage-side spiral grooves **151***a* to **151***c* therebetween. A radially outer end of the second portion **317***b* does not communicate with the outer space S**2**.
(72) The third portion **317***c* is a portion extending concentrically with the stationary seal ring **300**, and connecting radially inner ends of the first portion **317***a* and the second portion **317***b*. The fourth portion **317***d* is a portion extending concentrically with the stationary seal ring **300**, and connecting radially outer ends of the first portion **317***a* and the second portion **317***b*.
(73) According to this configuration, since the endless deep groove **317** is disposed between a leakage-side region A**20**′ and the outer space S**2**, the amount of the sealed fluid F flowing into the leakage-side region A**20**′ is small, so that the leakage of the sealed fluid F into the inner space S**1** can be reduced as much as possible.
Fifth Embodiment
(74) Next, a sliding component according to a fifth embodiment of the present invention will be described with reference to FIG. **8**. Incidentally, the description of configurations that are the same as and overlap with the configurations of the second embodiment will be omitted.
(75) A mechanical seal of the fifth embodiment is an outside mechanical seal that seals the sealed fluid F in the inner space S**1** and that allows the outer space S**2** to communicate with the atmosphere A. Namely, in the fifth embodiment, the inner space S**1** functions as a fluid-side space, and the outer space S**2** functions as a leakage-side space.
(76) An annular land **18**′ is provided at an edge portion on the radially outer side of a sliding surface **411** of a stationary seal ring **400** of the fifth embodiment. An annular deep groove **117**′ as a communication groove is formed on the radially inner side of the annular land **18**′. Four deep grooves **417**, each communicating with the inner space S**1** at both ends, are provided in the circumferential direction on the radially inner side of the annular deep groove **117**′. The deep grooves **417** do not communicate with the outer space S**2**.
(77) A fluid-side spiral groove **413** and a fluid-side reverse spiral groove **414** are provided in a fluid-side region A**100**. Leakage-side spiral grooves **451***a* to **451***c* and leakage-side reverse spiral grooves **461***a* to **461***c* are provided in a leakage-side region A**200**. The leakage-side spiral grooves **451***a* to **451***c* and the leakage-side reverse spiral grooves **461***a* to **461***c* communicate with the annular deep groove **117**′, and extend to the radially inner side.
(78) The embodiments of the present invention have been described above with reference to the drawings; however, the specific configurations are not limited to the embodiments, and the present invention also includes changes or additions that are made without departing from the scope of the present invention.
(79) For example, in the first to fifth embodiments, the mechanical seal has been described as an example of the sliding component; however, the present invention may be applied to other mechanical seals for general industrial machines, automobiles, water pumps, and the like. In addition, the present invention is not limited to the mechanical seal, and may be applied to sliding components other than the mechanical seal, such as a slide bearing.
(80) In addition, in the first to fifth embodiments, the mode in which the fluid-side grooves and the fluid-side reverse grooves directly communicate with the fluid-side space has been provided as an example; however, as in a stationary seal ring **500** illustrated in FIG. **9**, a fluid-side groove **513** may extend from the first portion **171** of the deep groove **17** in the one circumferential direction so as to be concentric with the stationary seal ring **500**, and a fluid-side reverse groove **514** may extend

from the second portion **172** of the deep groove **17** in the other circumferential direction so as to be concentric with the stationary seal ring **500**. In other words, the fluid-side groove **513** and the fluid-side reverse groove **514** may communicate with the outer space S**2**, namely, the fluid-side space via the deep groove **17**. In addition, the fluid-side groove **513** and the fluid-side reverse groove **514** may not have a radial component, namely, may not be spiral grooves.
(81) In addition, in the first to fifth embodiments, the mode in which the fluid-side groove and the fluid-side reverse groove have a symmetrical shape, the leakage-side groove and the leakage-side reverse groove have a symmetrical shape, and a positive pressure and a relative negative pressure are generated in the same manner in both the forward direction and the reverse direction of relative rotation has been provided as an example; the present invention is not limited to the mode, and each groove may be formed in an asymmetrical shape, so that a positive pressure and a relative negative pressure are generated differently between the forward direction and the reverse direction of relative rotation.
(82) In addition, in the first to fifth embodiments, the mode in which the communication groove communicating with the leakage-side end portions of the leakage-side grooves and the leakage-side reverse grooves is a deep groove has been provided as an example; however, the communication groove may be a shallow groove having such a depth that a dynamic pressure is generated by relative rotation of the mechanical seal.
(83) In addition, in the second to fifth embodiments, the mode in which the communication groove has an annular shape has been provided as an example; however, the present invention is not limited to the mode, and the communication groove may extend in an arcuate shape to communicate with the leakage-side end portions of the leakage-side grooves and the leakage-side reverse grooves. Incidentally, the communication groove is not limited to an arcuate shape, and may extend in a sinusoidal shape or a linear shape.
(84) In addition, the number of the grooves and the deep grooves may be freely changed.
(85) In addition, in the first to fifth embodiments, the mode in which the degrees of inclination of the fluid-side groove and the leakage-side reverse groove and of the fluid-side reverse groove and the leakage-side groove in the circumferential direction as viewed in the axial direction are almost the same has been provided as an example; however, the degrees of inclination in the circumferential direction may be different from each other.
(86) In addition, in the first to fifth embodiments, the mode in which each groove is provided on the stationary seal ring has been provided as an example; however, each groove may be provided on the rotating seal ring.
(87) In addition, in the first to fifth embodiments, the mode in which each groove has a constant depth in the extension direction has been provided as an example; however, a step or an inclined surface may be provided on a bottom surface of each groove.
(88) In addition, in the first to fifth embodiments, the mode in which the fluid-side groove and the fluid-side reverse groove are shallower than the leakage-side groove and the leakage-side reverse groove and have a smaller capacity has been provided as an example; however, the depth and capacity of the fluid-side groove and the fluid-side reverse groove may be formed to be approximately the same as the depth and capacity of the leakage-side groove and the leakage-side reverse groove.
(89) In addition, the sealed fluid side and the leakage side have been described as a high-pressure side and a low-pressure side, respectively; however, the sealed fluid side and the leakage side may be a low-pressure side and a high-pressure side, respectively, or the sealed fluid side and the leakage side may have substantially the same pressure.
(90) In addition, in the first to fifth embodiments, the sealed fluid F has been described as a high-pressure liquid, but is not limited thereto, and may be a gas or a low-pressure liquid or may be in the form of a mist that is a mixture of liquid and gas.
(91) In addition, in the first to fifth embodiments, the leakage-side fluid has been described as the

atmosphere A that is a low-pressure gas, but is not limited thereto, and may be a liquid or a high-pressure gas or may be in the form of a mist that is a mixture of liquid and gas.

REFERENCE SIGNS LIST

(92) **10** Stationary seal ring **11** Sliding surface **12** Land **13** Fluid-side spiral groove (fluid-side groove) **13**A End portion **13**B End portion (terminating end portion) **14** Fluid-side reverse spiral groove (fluid-side reverse groove) **14**A End portion **14**B End portion (terminating end portion) **15**A End portion **15**B End portion (terminating end portion, leakage-side end portion) **15***a* to **15***c* Leakage-side spiral groove (leakage-side groove) **16**A End portion **16**B End portion (terminating end portion) **16***a* to **16***c* Leakage-side reverse spiral groove (leakage-side reverse groove) **17** Deep groove **18** Annular land **20** Rotating seal ring **21** Sliding surface **117** Annular deep groove A Atmosphere A**1** Fluid-side region A**2** Leakage-side region F Sealed fluid S**1** Inner space (leakage-side space) S**2** Outer space (fluid-side space)

Claims

1. A sliding component, comprising a pair of sliding surfaces for being disposed in a rotating machine to face each other at a location where the pair of sliding surfaces rotate relative to each other when the rotating machine is driven, and partitioning a fluid-side space and a leakage-side space off from each other, wherein one of the sliding surfaces is provided with: a fluid-side groove which has an opened end portion communicating with the fluid-side space and a closed end portion opposed to the opened end portion in a radial direction and which extends from the opened end portion to the closed end portion in a forward direction of relative rotation of the sliding surfaces; a fluid-side reverse groove which has an opened end portion communicating with the fluid-side space and a closed end portion opposed to the opened end portion in a radial direction and which extends from the opened end portion to the closed end portion in a reverse direction of the relative rotation; a leakage-side groove of which at least one end portion is disposed closer to a side of the leakage-side space than the fluid-side groove and the fluid-side reverse groove, and which extends from the end portion toward the fluid-side space in the forward direction of the relative rotation; a leakage-side reverse groove of which at least one end portion is disposed closer to the side of the leakage-side space than the fluid-side groove and the fluid-side reverse groove, and which extends from the end portion toward the fluid-side space in the reverse direction of the relative rotation; and a deep groove that partitions a fluid-side region where the fluid-side groove and the fluid-side reverse groove are provided and a leakage-side region where the leakage-side groove and the leakage-side reverse groove are provided off from each other, and the fluid-side reverse groove is disposed on a downstream side in the forward direction of relative rotation with respect to the fluid-side groove in the fluid-side region.
2. The sliding component according to claim 1, wherein the deep groove communicates with the fluid-side space.
3. The sliding component according to claim 1, wherein each of the leakage-side end portions of the leakage-side groove and the leakage-side reverse groove communicate with the leakage-side space.
4. The sliding component according to claim 1, wherein the one of the sliding surfaces is provided with a communication groove communicating with each of leakage-side end portions of the leakage-side groove and the leakage-side reverse groove.
5. The sliding component according to claim 4, wherein the communication groove has an annular shape.
6. The sliding component according to claim 1, wherein each of terminating end portions of the leakage-side groove and the leakage-side reverse groove is disposed closer to a fluid-side space side than each of terminating end portions of the fluid-side groove and the fluid-side reverse groove.

7. The sliding component according to claim 6, wherein the fluid-side groove, the leakage-side reverse groove, the fluid-side reverse groove, and the leakage-side groove are disposed in an inclined manner, and the deep groove includes a first inclined portion extending along the fluid-side groove and the leakage-side reverse groove, and a second inclined portion extending along the fluid-side reverse groove and the leakage-side groove.

8. The sliding component according to claim 2, wherein each of the leakage-side end portions of the leakage-side groove and the leakage-side reverse groove communicate with the leakage-side space.

9. The sliding component according to claim 2, wherein the one of the sliding surfaces is provided with a communication groove communicating with each of leakage-side end portions of the leakage-side groove and the leakage-side reverse groove.

10. The sliding component according to claim 9, wherein the communication groove has an annular shape.

11. The sliding component according to claim 1, wherein the deep groove is an annular deep groove which does not communicate with the fluid-side space and the leakage-side space.

12. The sliding component according to claim 1, wherein deep groove has fluid-side deep groove portions which are disposed to be closer to the fluid-side space than leakage-side groove and the leakage-side reverse groove and which extend in the circumferential direction, leakage-side deep groove portions which are disposed to be closer to the leakage-side space than the fluid-side groove and which extend in the circumferential direction, the fluid-side reverse groove and the fluid-side deep groove portion, and slant deep groove portions each of which connect the fluid-side deep groove portion and the leakage-side deep groove portion adjacent to each other in the circumferential direction, the leakage-side region is between the fluid-side deep groove portion and the leakage-side space in the radial direction, and the fluid-side region is between the leakage-side deep groove portion and the fluid-side space in the radial direction.